
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

Anonymous authors
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ABSTRACT

Mixture of Experts (MoE) models have become a central mechanism for scaling large language models, yet their internal feature geometry is still less mechanistically characterized than that of dense networks. Prior toy-model work explains dense-model *superposition* as a response to feature sparsity and feature importance: dense networks represent more features than dimensions by packing sparse features into shared directions. We show, in a controlled MoE extension of this setting, that the dense-model account does not transfer unchanged to routed experts. Feature sparsity and feature importance do not produce the same discontinuous phase changes observed in dense models; instead, MoE behavior is more directly organized by network sparsity, defined by the ratio of active to total experts. We introduce expert-aware measures of superposition and find that greater network sparsity is associated with greater *monosemanticity* and reduced global interference. We also formulate expert specialization as a representational property: an expert specializes when it both occupies feature directions and represents the corresponding features monosemantically. Under appropriate initialization, experts organize around coherent feature combinations rather than merely satisfying load-balancing criteria. These results establish network sparsity as a concrete mechanism by which MoEs can alter superposition and improve interpretability-relevant structure while retaining comparable reconstruction performance in the studied toy reconstruction settings.

1 INTRODUCTION

Mixture of Experts (MoEs) are now a prominent design principle in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025). Their appeal is architectural: by activating only a subset of parameters on each input, they improve the efficiency–performance trade-off relative to fully dense computation (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work has refined the routing problem, for example through Expert-Choice routing, and has improved training stability and transfer, as in ST-MoE (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical analyses further show that learnable routers can recover latent cluster structure in data, providing one explanation for the emergence of expert specialization (Dikkala et al., 2023). Despite this progress, the representational mechanisms induced by routing remain comparatively underexplained from the standpoint of mechanistic interpretability.

Interpretability-oriented MoE methods have begun to make expert behavior more transparent. Yang et al. (2025b) propose MoE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity; Park et al. (2025) introduce MONET, scaling the number of experts to support capability editing through expert activation. These approaches demonstrate that expert-level structure can be used for interpretability interventions. They do not, however, by themselves explain the feature geometry that routed experts learn. The central mechanistic questions therefore remain open: how do MoEs represent features, how does routing reshape superposition, and when does specialization emerge without being imposed by an explicit semantic objective? Consistent with this gap, Mu & Lin (2025) identify mechanistic interpretability as a key open challenge for MoE research.

A fundamental obstacle to interpreting neural networks is *superposition*: the regime in which a model represents more features than it has available dimensions. Superposition allows a dense network to store many sparse features in a limited representational space, but the resulting non-orthogonal packing makes individual neurons polysemantic and weakens the correspondence between a neuron and a human-interpretable feature.

MoE architectures introduce a distinct variable into this problem: network sparsity. Unlike dense models, which apply the same parameter set to every input, MoEs activate only a fraction of their total parameters (Shazeer et al., 2017). Conditional computation therefore changes the representational trade-off. A dense model exploits feature sparsity by packing rarely co-active features into shared dimensions; an MoE can instead route inputs to different experts and localize interference within expert-specific subspaces. The relevant mechanistic question is therefore not only whether input features are sparse, but how routing-induced network sparsity changes the geometry of feature representation.

We study this question through three linked investigations. First, we ask whether MoEs exhibit less superposition than dense models under matched parameter conditions. Second, we test whether MoE experts display the discrete phase changes in feature representation that dense toy models exhibit as feature sparsity and relative importance vary. Third, we examine whether expert specialization can be characterized by feature representation itself, rather than by aggregate load-balancing statistics alone.

We answer these questions by extending the toy-model framework of Elhage et al. (2022) to MoEs. The paper makes three contributions. First, it identifies network sparsity as a central variable for MoE superposition: as the number of experts increases under the studied settings, MoEs exhibit more monosemantic representations and less feature interference than dense models with comparable parameter budgets. Second, it shows that MoEs do not reproduce the sharp dense-model phase changes induced by feature sparsity and feature importance; their transitions are more continuous and expert-specific. Third, it formulates expert specialization as a feature-representational property—the joint occupancy and monosemantic representation of feature directions—and shows that, when initialized appropriately, experts organize around coherent feature combinations rather than merely satisfying load-balancing criteria.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis proposes that networks encode concepts as directions in activation space (Park et al., 2024), but the number of interpretable features can exceed the dimensionality available to represent them. Elhage et al. (2022) formalized this mismatch as superposition, showing that dense models use non-orthogonal feature packing to increase representational capacity at the cost of polysemanticity. Although superposition can be capacity-efficient (Scherlis et al., 2025), it also motivates post-hoc disentanglement methods, including Sparse Autoencoders, that attempt to recover monosemantic features from polysemantic representations (Bricken et al., 2023). Recent work further characterizes how data correlations shape superposition (Prieto et al., 2025) and how interference patterns arise and may be mitigated (Gurnee et al., 2023). We extend this line of work from dense architectures to MoEs and show that network sparsity, not feature sparsity alone, is necessary to characterize the representational strategies of routed expert models.

Interpretability of MoEs. MoEs are a leading architecture for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), yet mechanistic accounts of their internal representations remain less developed than analogous accounts for dense networks. Existing analyses often emphasize macroscopic behavior, including routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). More recent interpretability-motivated approaches encourage specialization through architectural constraints, including predefined concept routing (Yang et al., 2025b) and expert-count scaling for capability editing (Park et al., 2025). These studies establish that expert structure can be useful, but they do not fully explain the intrinsic feature geometry learned by MoEs. Our contribution is to analyze how experts

alter superposition directly, to show that MoEs exhibit greater monosemanticity than dense models in the studied toy settings, and to define specialization in terms of feature occupancy and monosemantic representation.

3 BACKGROUND

A central objective of mechanistic interpretability is to reverse engineer neural networks by decomposing their internal representations into human-interpretable concepts, commonly called ‘features’. In the linear feature view, a hidden state h is represented as a superposition of feature directions,

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ denotes the direction associated with feature i , α_i denotes the feature’s activation strength, and F is the set of represented features. Under this interpretation, α_i approximates the degree to which feature i is present in the input.

The *superposition hypothesis* states that models can represent substantially more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Achieving this compression requires feature vectors in F to be packed into a latent space in which they are not all mutually orthogonal. When two features have nonzero interference, $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say that they are in *superposition*. Such interference can be tolerable when features are sparsely active, as is often assumed for text-derived features, but it reduces interpretability because α_i may include spurious activation attributable to features other than i .

Monosemanticity is typically defined at the level of individual neurons: a neuron’s activation is monosemantic when it cleanly corresponds to a single α_i , or equivalently when the relevant feature is basis-aligned. Orthogonal features that are not basis-aligned can still yield polysemantic neurons even when feature interference is small. Our analysis therefore focuses on reducing superposition rather than enforcing basis alignment. For brevity, when we describe models, experts, or features as “more monosemantic,” we mean that they exhibit less superposition under the measurements defined below.

4 DEMONSTRATING SUPERPOSITION

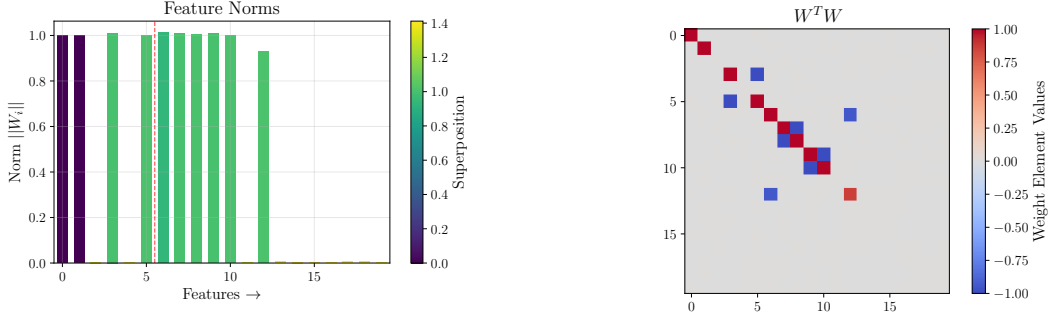
MoEs are often described as compositions of dense subnetworks, with each expert treated as an independent dense model selected by a router. That description is architecturally accurate but mechanistically incomplete: it does not determine whether experts represent features in the same geometric manner as dense models. We therefore compare dense and MoE models directly in terms of feature representation strength and superposition.

4.1 EXPERIMENTAL SETUP

Our experimental objective is to determine how an MoE projects a high-dimensional feature vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and reconstructs the original input. This construction extends the toy superposition framework of Elhage et al. (2022) from dense autoencoders to routed expert architectures.

The input distribution. The input vector x represents activations in an idealized disentangled representation: each coordinate x_i is a distinct independent feature, corresponding to a perfectly neuron-aligned and monosemantic basis. We sample nonzero feature values as $x_i \sim U(0, 1)$ and impose sparsity $S \in [0, 1)$ by setting $P(x_i = 0) = S$. To model unequal feature utility, each feature x_i receives a scalar importance I_i that weights its contribution to reconstruction loss. For phase-change experiments, we isolate the last feature by scaling its relative importance by $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. The parameters I_i and r therefore control feature signal strength while preserving the same underlying sparse input structure.

Model Architecture. The MoE contains E experts. Each expert e is parameterized by a projection matrix $W^e \in \mathbb{R}^{m \times n}$ and a reconstruction bias $b^e \in \mathbb{R}^n$. A learned router



(a) Norm of each feature’s weight vector $\|W_i\|$; colors indicate whether the feature is represented in superposition (green) or more monosemantically (purple).

(b) $W^T W$ matrix in which each cell reports $(\hat{W}_i \cdot W_j)$, making pairwise feature interference visible.

Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, importance $I = 0.7^i$, and uniform feature density $(1 - S) = 0.1$. Superposition (color) is computed as $\sum_j (\hat{W}_i \cdot W_j)^2$.

$g(x) = \text{softmax}(W^r x)$, with $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Each active expert maps the input to a lower-dimensional hidden state $h^e = W^e x$ and produces a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^T h^e + b^e)$ ¹. The model output is the weighted sum of active-expert reconstructions, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where w_e denotes the renormalized gating weight.

We train all models using an L2 reconstruction objective weighted by feature importance, I_i , with $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, we add the standard auxiliary load-balancing loss (Fedus et al., 2022) to prevent expert collapse, defined as $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the total number of experts, f_e is the fraction of samples in a batch routed to expert e , P_e is the average gating probability assigned to expert e across the batch, and $\alpha = 0.01$ sets the penalty strength. We deliberately omit auxiliary load balancing in Section 4 so that the comparison isolates the intrinsic architectural bias of MoEs with respect to superposition.

4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we measure two complementary properties: *representation strength* and *interference*. Representation strength is the norm of feature i ’s weight vector in expert e , $\|W_i^e\|$. Values near one indicate that feature i is represented in expert e , whereas values near zero indicate that the feature is not learned by that expert. Interference is visualized through the Gram matrix $W^T W$, whose off-diagonal entries quantify pairwise interactions among feature directions.

Figures 1a and 2a show that, under equal total hidden dimensionality ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$), the dense model and the MoE represent a comparable number of features (10 versus 8) with similar feature norms. The difference lies in the structure of interference. Although individual MoE experts can exhibit local superposition, as in Expert 0 of Fig 2b, the global interference pattern is partitioned by routing. In the dense model (Figure 1b), any represented feature can interfere with any other; in the MoE, features routed to different experts have zero cross-expert interference, producing a block-structured pattern. This comparison shows that MoEs allocate capacity by partitioning the feature space, thereby reducing the global scope over which interference can occur.

¹The output ReLU enforces non-negative reconstructions, matching the input distribution in which $x_i \in [0, 1]$ when nonzero. The off-diagonal terms in $(W^e)^T W^e$ can create negative interference, and the ReLU suppresses these negative components. When features are sparse, such negative interference is effectively filtered to zero, favoring configurations with negative off-diagonal terms, such as antipodal pairs.

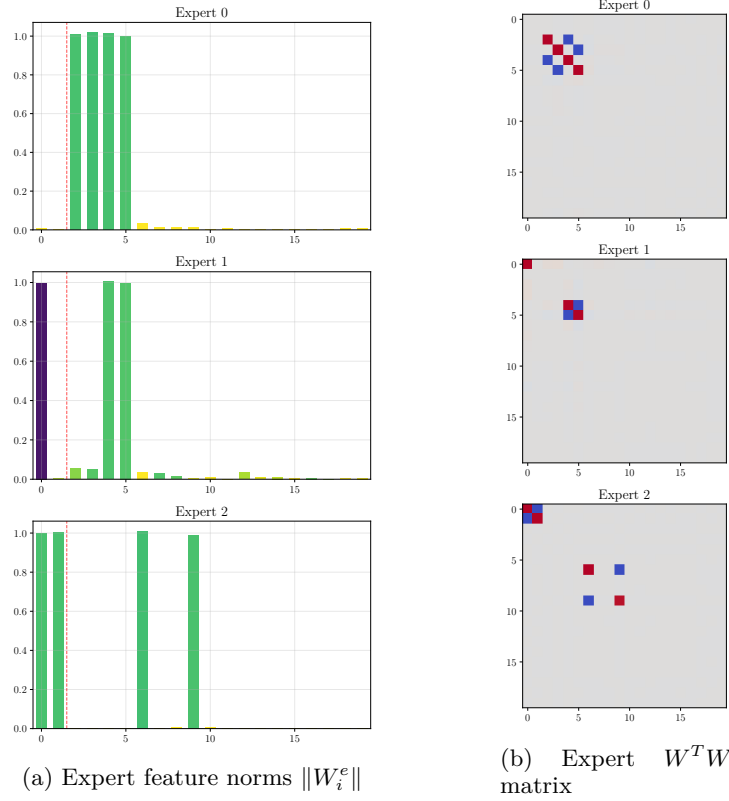


Figure 2: Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under top- $k = 1$ routing, with importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

Expert Feature Dimensionality. To quantify how MoEs allocate limited representational capacity relative to dense models, we measure feature dimensionality: the “fraction of a dimension” assigned to a given feature (Elhage et al., 2022). For feature i in expert e , dimensionality is defined as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e ranges from zero, indicating that the feature is not learned, to one, indicating a monosemantic representation. The aggregate MoE capacity is therefore $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. A representation is efficiently packed when the sum of feature dimensionalities approximately equals the number of embedding dimensions, $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For MoEs, the analogous relation sums over experts: $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, indicating that MoEs preserve the same feature-packing efficiency as dense models with the same total parameter budget while altering the distribution of interference.

Features per Dimension & Network Sparsity. Because dense and MoE models both efficiently pack features, the key comparison is not whether capacity is used, but how many features are represented per active dimension. Features per dimension provides a direct measure of superposition: values greater than one imply that the model represents more features than available dimensions. For MoEs, we define this quantity as

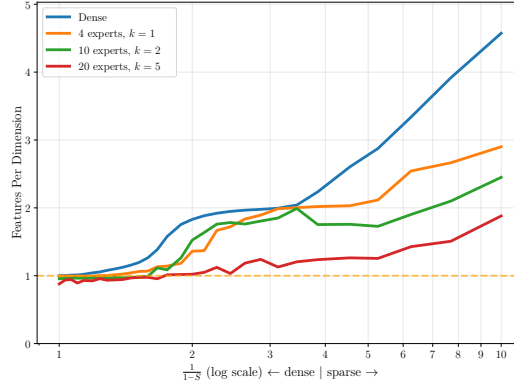


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-g}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the greatest superposition. Superposition decreases as expert count increases: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameters and similar k/E . The dashed line at 1.0 marks the monosemantic threshold.

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the Frobenius norm and p_e is the expected usage probability of expert e over a batch, computed as the average renormalized gating weight after top- k routing.

For MoEs with the *same* total parameter count as the dense model, Figure 3 shows that the dense model has more features per dimension and therefore more superposition. The dense model uses superposition to represent more learned features than total dimensions ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE tends to remain near the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Increasing the number of experts while keeping total parameters and the ratio k/E approximately fixed further reduces features per dimension. Thus, in these experiments, greater expert count—and hence greater network sparsity—corresponds to less superposition and more monosemantic feature representation. The dense model’s greater use of superposition yields consistently lower reconstruction loss than the MoEs in Figure 6 of Appendix A.1, with a loss difference of $\sim 0.03 - 0.08$ at a given sparsity level. Nevertheless, as the number of experts increases, MoE loss becomes consistently closer to dense-model loss, indicating that network sparsity improves interpretability-relevant structure while preserving comparable reconstruction quality in these toy settings. Appendix A.2 provides theoretical intuition for this pattern.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute them across experts with less interference. This suggests that network sparsity reshapes how features are allocated rather than how many are learned. We examined how properties of the input distribution—such as feature sparsity and importance—drive this allocation and whether they induce physics-inspired *phase changes* in representation.

Models have a finite way of representing features; each feature may be ignored, superimposed, or monosemantic. Phase change is the observation that sometimes there are discrete boundaries between regions, which are functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). By varying the sparsity and relative importance of features in

the input distribution, we can elicit different behavior; for example, more feature sparsity encourages greater superposition. Analyzing the phase diagram of each expert in MoEs demonstrates they employ different representational strategies compared to dense models.

Setup. We follow the same setup as Section 4.1, except with load balancing loss to avoid specialization collapse—when experts and routers fall into local minima where certain experts are entirely ignored—in certain setups. There are three models setups, all with one active expert ($k = 1$): (A) $n=2, m=1$; (B) $n=3, m=1$; and (C) $n=3, m=2$. We report the expert-specific phase diagram across feature sparsity and last-feature relative importance for varying network sparsity by increasing the experts (E) up to the number of input feature dimensions (n).

In this section we fix active parameters rather than total parameters such that $m_{dense} = km_{moe}$ (ignoring router parameters). The reason is to compare within model architectures; otherwise, any observed differences could be attributed to architectural changes instead of the number of active parameters. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. But the latter has only one hidden dimension ($m = 1$) to encode the same number of input features ($n = 3$) as the former, with two hidden dimensions ($m = 2$), making it difficult to use superposition to understand specialization.

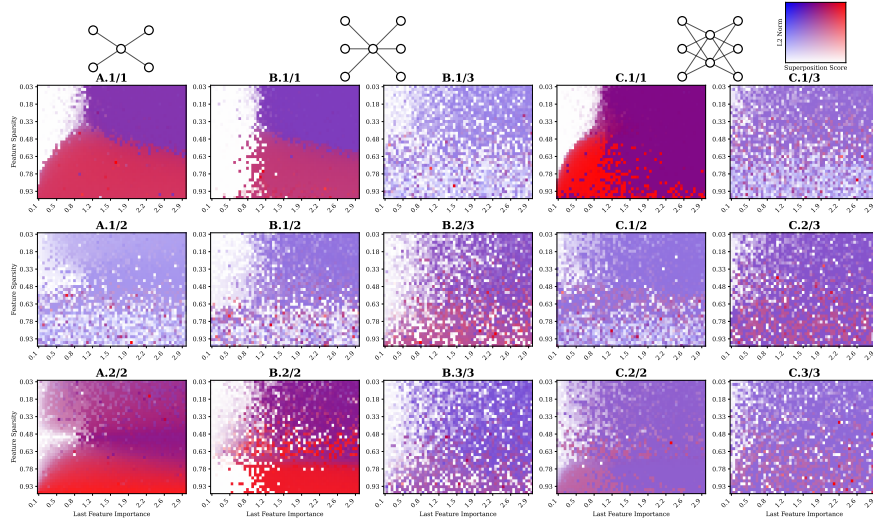


Figure 4: For a particular expert and input dimension (feature), we can decode how it is embedded in the hidden dimension—whether it is ignored (white), monosemantic (blue-purple), or superimposed (red). We plot joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) across varying feature sparsity $S \in [0.1, 1]$ and relative last feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last feature of n total features. For each cell, we train ten models and select the one with the lowest loss. We used load balancing loss in this section. We plot joint feature norm and superposition for the last feature: low L2 norm ($\|W_n\|$) is white, denoting the model is ignoring the last feature; otherwise a low superposition score is blue-purple to indicate monosemantic representation of the last feature. Red indicates the feature is represented in superposition. Cell (i, j) in subfigure X.e/E denotes the expert e of E total experts trained on architecture X for last feature importance $I_n = i$ and sparsity $S = j$; X.1/1 indicates a dense model.

Results and Takeaways. In all single-expert (dense) cases, we observed a clear phase change (Figure 4.X.1/1), affirming the work of Elhage et al. (2022). When we increased the total number of experts, discrete phase changes disappeared. Some experts in MoEs with $E = 2$ are reminiscent of their respective dense cases (Figure 4.X.2/2), but exhibit more continuous transitions. In each case, one expert became more monosemantic, specializing in the most important feature by relative importance. Experts dissimilar from the dense cases universally have much lower superposition scores (they are bluer), indicating more

monosemantic representations. This aligns with the conclusions of the previous section—MoEs favor lower superposition scores compared to their dense counterparts.

For the $n = 2, m = 1$ setup (Figure 4.A), the dense model does not represent the last feature when feature sparsity is low. However, the comparable MoE model preserves the last feature much more because it has the capacity. With three input dimensions (Figure 4.B), the MoEs do not exhibit this behavior because the experts are superimposing the other two features; there is no space for the third feature within one hidden dimension. Unlike the other two cases, for architecture B the hidden dimension with superposition is not sufficient, in the high-sparsity regime, to represent all features. Yet we do not see clear phase change—except for the 0.5 – 0.7 feature sparsity region in 4.B.2/2, where it is mostly discrete but mixed. For $m = 2$ the white region in the dense model (Figure 4.C.1/1) (in the mid- to low-feature sparsity domain, when the feature is relatively less importance than the others) ignores the last feature. However, as network sparsity increases—across all other Figure 4.C—the models represent the last feature with greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). In other words, the dimensionality in the low relative-importance region increased with increasing network sparsity, as demonstrated in Figure 3.

We observed a window of feature sparsity from roughly 0.48 to 0.7 in Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 where there is heavy mix of polysemanticity, monosemanticity, or ignorance. This indicates there is a middleground in MoEs with comparable loss between polysemantic and monosemantic representations which make it difficult to consistently commit to the strategies we observe in low and high feature sparsity domains. We see no such pattern in dense models—evidence that MoEs learn different representational strategies.

Conclusion. These experiments are all top- $k = 1$, so only one expert is active at a time. Even so, we see vastly different behavior even the in $E = 2$ case, including when the hidden dimension capacity with superposition is sufficient to represent all features. This leads us to conclude it is misleading to think of MoEs as an aggregation of dense models. The mechanism of the router which allows experts to observe only a subset of the feature domain vastly modifies the behavior and learning of the experts.

6 EXPERT SPECIALIZATION

Since MoEs exhibit less superposition, we now examine the organization of such monosemantic features within experts and its relation to specialization.

Expert specialization in MoEs traditionally centers around load balancing between experts across all inputs (Chaudhari et al., 2025). However, this fails to capture the natural intuition of specialization, wherein an expert is only used when appropriate concepts—those the expert is specialized in—are present in the input.

We define an expert as specialized if it *occupies* certain feature directions in the input space, and if it represents said features relatively monosemantically. We demonstrate that these two conditions are directly correlated, and show how the presence of these two conditions encourages load balancing across experts.

Because we fix $k = 1$, the feature space is partitioned into convex cone regions (see Appendix A.3), with each region routed to a particular expert. By definition, this means $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained within the cone and s is any positive scalar. If a particular feature vector $\vec{x}$ is routed to an expert, then all $s\vec{x}$ are routed to that same expert. In this case, we say that feature x is contained within expert e , and as such expert e *occupies* x .

We empirically find that small models that distribute the input space across more experts tend to achieve lower loss (see Appendix A.4). This warrants a question: does the allocation of the input space to certain experts imply any characteristics regarding those experts? Our definition of expert specialization suggests that this allocation implies *monosemanticity*, which we will see is a correlation that holds for larger toy models (e.g., $m = 10$).

In models with $n > 2$, we explore whether initializing experts to occupy features in the input space cause the experts to be more monosemantic w.r.t. those features. Separately,

we see if, for the features an expert has chosen to represent monosemantically, the expert occupies those features in the input space.

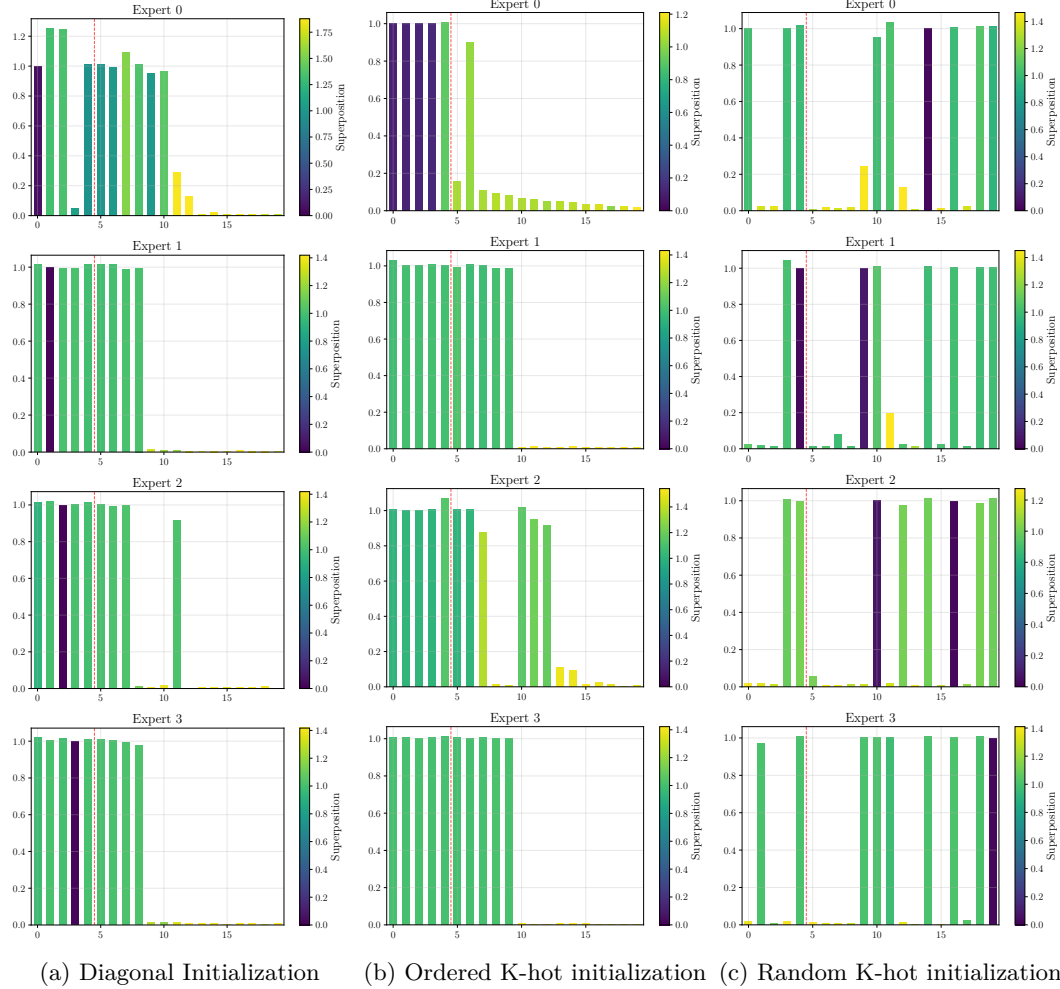


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) results for three different initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), and relative feature importance decreases exponentially in order from feature one to 20. In (b), the gate matrix is initialized to an “ordered k-hot”, such that the first expert aligns with the first five features, and each subsequent expert aligns with the next five features. Relative feature importance is the same as (a). In (c), the gate matrix is initialized to a “random k-hot”, where each expert is assigned five random features such that experts share no common feature but cover all 20 features collectively. Relative feature importance decreases exponentially but is randomly distributed across features.

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied, and only that feature, as shown in Figure 5a. When the router is ordered k-hot initialized, the first expert monosemantically represents four of the five features it initially occupied, as shown in Figure 5b. The other experts, initialized over other features, did not monosemantically represent these less important features, nor did they monosemantically represent the five most important features they were not initialized over. When we break the ordering of feature importance and randomize the features each expert initially occupies, each expert monosemantically represented only the most important feature(s) it was initialized over, as shown in Figure 5c.

There is a strong correlation between the features that are initially routed to an expert and which features that expert represents monosemantically. Furthermore, we observe that experts only monosemantically represent important features. This is true if we initialize each expert with one important feature explicitly, or if we give it a set of features, upon which it selects the most important feature itself and represents it monosemantically.

In the case of uniform feature importance, experts place all features in superposition with higher levels of interference. Despite this, the features an expert initially occupies are still *relatively* more monosemantic, on average achieving superposition scores 1.03 standard deviations below the average for that expert.

We investigated whether there is a correlation between experts representing certain features monosemantically, and said experts occupying those features in the input. To do this, we measure usage statistics when those features are *one of many* active features, and when they are the *only* active features. This second case is equivalent to measuring the probability that the expert occupies these features. The correlation holds both in xavier and k-hot initialization schemes, as seen in Table 1. Given $E = 10$, a mean expert usage of $\sim 10\%$ indicates an even load balancing across experts. In all cases, when the corresponding monosemantic feature(s) for an expert is active, the usage of the expert increases significantly. When this feature(s) is the only active feature, the expert dominates the usage. In the k-hot initialization scheme, 100% of all features monosemantically represented by an expert are occupied by that same expert.

Table 1: Monosemantic feature and usage statistics per expert for $n = 100, m = 10, E = 10$. One hundred models are trained for each initialization scheme (xavier and k-hot), providing 1000 experts in total for each. Each statistic is aggregated across models, classifying experts based on the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we track the expert usage when said feature(s) is one of several active features in the input, as well as the expert usage when said feature(s) is the *only* active feature in the input.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

As experts represent more features monosemantically, they can be seen as more specialized. Their usage on arbitrary input decreases, but conditional on their specialized features being active, their usage increases far greater than other experts. This holds true for all cases except the xavier initialized model with a four monosemantic feature expert, where there is a significant drop in utilization.

7 CONCLUSION

We investigated how experts affect superposition in MoEs, showing that MoEs consistently exhibit greater monosemanticity than dense networks while not exhibiting a phase change. We proposed a feature-based definition of expert specialization, demonstrating that experts naturally organize around coherent features when initialization encourages this specialization. However, our findings are based on simple autoencoder toy models with synthetic data, leaving open questions about generalization to large-scale transformers where the feature distribution is unknown (see Appendix A.6). Despite these limitations, we show how toy MoEs achieve comparable loss while maintaining more interpretable representations—challenging the prevalent zeitgeist that mechanistic interpretability and model capability are fundamentally in tension. Future work should explore what favors monosemanticity in MoEs, how training dynamics of MoEs differ from those of the dense model, and when specialization emerges. Answering these questions can inform the design of more interpretable, high-performing language models.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

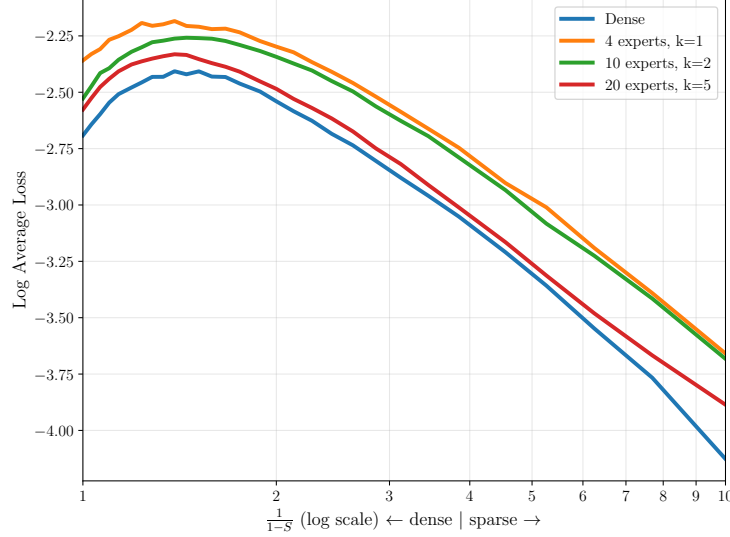


Figure 6: Log average loss versus feature density ($\frac{1}{1-s}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. Although dense model outperforms all MoEs at every sparsity level, as the number of experts increases, the MoE loss gets closer to the dense model.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, however, “eat this same gap” structurally through conditional computation. By aligning activation sparsity with feature sparsity, MoEs remove the computational penalty for having dedicated, rarely-activating neurons. As noted by Elhage et al. (2022), when a model only expends computation on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy, effectively trading the *compression* of superposition for the *selection* of routing.

The structural change manifests geometrically as a reduction in *interference*. While the global ratio of number of features represented to parameters remains constant, the router effectively partitions the feature space, ensuring that a feature routed to expert e only competes for capacity with the subset of features also assigned to that expert. Consequently, interference is governed by the local expert matrix $(W^e)^\top W^e$ rather than the global $W^\top W$ of a dense model. This partitioning drastically reduces the number of interfering features for any given feature vector, minimizing the optimization pressure to pack features in superposition and allowing experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the regime of $k = 1$, the router function $g(x) = \text{softmax}(W^r x)$ is equivalent to $\text{argmax}(W^r x)$. The region routed to expert i can be represented as $\forall j \neq i, (w_i - w_j)^\top x > 0$ where w_i and w_j are row vectors of W^r . This is a homogeneous linear inequality. Regions bounded by such inequalities are by definition convex cones. If a particular x satisfies this inequality, then multiplying both sides by any positive scalar s will still satisfy the inequality.

ity. Furthermore, if x_1 and x_2 satisfy this inequality, then any $x = x_1\lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$ will also satisfy the inequality.

In the case of $k > 1$, the region of inputs which get sent to a particular expert e becomes a union of convex cones. Generally, the union of a convex cone is not itself a convex cone. Therefore, the understanding of experts occupying feature directions may not hold beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

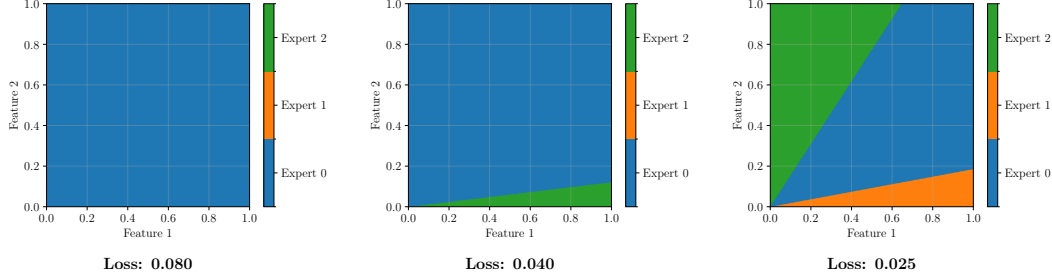


Figure 7: Expert routing of three identical models with differing initialization schemes. We use $n = 2$, $E = 3$, $m = 1$. The first model (left) has the worst performance (loss: 0.08) and routes all inputs to one expert. The second model (middle) has better performance (loss: 0.04) and routes a small portion of inputs, specifically those when feature 1 is active, to a second expert. The third model (right) has the lowest loss (loss: 0.025), and distributes the input space among all experts. One expert is chosen when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2$, $m = 1$, $E > 1$), we empirically find that models that distribute the input space across more experts tend to achieve lower loss, testing with $E \in [2, 7]$. Holding $n = 2$ allows us to visualize which portions of the input space get routed to which experts, as seen in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map between any dense model and a monosemantic MoE with an equivalent number of active features under a sparsity constraint.

Assume there exists an upper bound for the number of active features a for any input such that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Furthermore, assume that a is no greater than the hidden dimensionality, m , of an expert, providing an upper bound on the number of features a model has to represent. Assume also that the hidden dimensions is smaller than the total number of input features n ($a \leq m \leq n$). To construct the monosemantic MoE, for each possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —meaning the size of the subset of active features is smaller than or equal to a —create an expert which monosemantically preserves those features. (In fact, you can take only the subsets such that $|S| = a$.) The router then selects the expert which corresponds to those active features (of which there will never be more than a , by assumption):

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has sufficient capacity to represent its assigned features without superposition. To reiterate, only a features are active and every unique combination of active features receives its own dedicated expert with sufficient capacity to represent those features monosemantically. So, the number of possible experts needed is $\binom{n}{m}$.

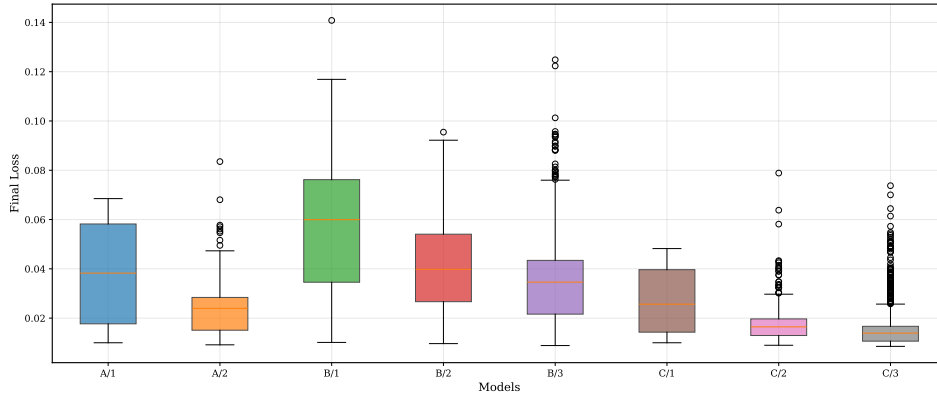


Figure 8: Model X/E uses X to denote the same model architectures and models used in Figure 4 and E denotes the total number of experts (i.e. network sparsity). Increasing network sparsity decreases mean loss while increasing localized variance—especially as the number of experts reaches the input feature dimensions. This can be attributed to the relatively unstable training of MoEs compared to dense models (despite training ten models for each cell and selecting the lowest loss).

The reconstruction for this theoretical MoE has zero loss only as the sparsity constraint holds (or goes to one in these toy models) because there is the chance more than m features could be active at one time ($a \not\leq m$), which would exceed the monosemantic representational capacity of the network (but the dense polysemantic could do no better unless features are correlated in the distribution). Therefore, even if $a \leq m$ sometimes, the polysemantic model encounters the same problem and the monosemantic MoE under this construction may still outperform it under looser sparsity constraints.

Thus, for any dense model, $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ under the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings described in this paper, the sparsity constraint holds in the limit where sparsity goes to one. However, in practice there may be an upper bound on the amount of features a particular amount of information can semantically encode, indicated by the size of meaningful embeddings of that data. Therefore, a MoE model with sufficient experts and a tractable amount of superposition (e.g. interpretable) may be sufficient to encode all features present.

A.6 LIMITATIONS

Our results should be interpreted in light of several limitations. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). While this enables precise measurement of interference and monosemanticity, it abstracts away many complexities of large-scale transformers, including multiple computation, attention, heterogeneous feature distributions, and realistic routing dynamics. Consequently, the transferability of our findings to real-world MoE architectures remains uncertain.

Second, our architectural and routing choices are intentionally restricted: we use simple experts, fixed top- k routing (often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs often employ richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022) or expanding beyond the task of reconstruction to next token prediction, for example, which may yield qualitatively different representational behaviors.

Third, the feature distributions in our synthetic tasks—including sparsity patterns, feature importance, and independence assumptions—are significantly simpler than those found in natural data.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work suggests such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, our study provides mechanistic insights under clean experimental conditions, but further work is required to validate these patterns in large, realistic MoE systems. Unfortunately, our methods do not scale naturally to larger models and this is a clear direction for future work in this space.